De Novo Mutations in the Sodium-Channel Gene SCN1A Cause Severe Myoclonic Epilepsy of Infancy

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Abstract

Severe myoclonic epilepsy of infancy (SMEI), or Dravet syndrome, is a rare disorder characterized by generalized tonic, clonic, and tonic-clonic seizures that are initially induced by fever and begin during the first year of life. Later, patients also manifest other seizure types, including absence, myoclonic, and simple and complex partial seizures, accompanied by psychomotor delay and ataxia.

Results / Case_Report

We screened the SCN1A gene (NM_001165963.4) in 7 unrelated probands diagnosed with Dravet syndrome. In Patient SMEI-1, we identified a de novo heterozygous nonsense mutation c.3733C>T (p.Arg1245Ter) in exon 19. Both parents tested negative for the variant, and paternity was genetically confirmed. Patient SMEI-1 presented at 5 months of age with prolonged hemiclonic febrile seizures, subsequently developing intractable myoclonic seizures, ataxia, and severe intellectual disability by age 3.

In Patient SMEI-2, direct sequencing revealed a de novo heterozygous missense substitution c.2216G>A (p.Arg739Gln) located within the highly conserved S4 transmembrane voltage-sensor segment of domain II. Neither parent carried the c.2216G>A alteration (paternity confirmed). Patient SMEI-2 exhibited febrile status epilepticus at 6 months, followed by generalized tonic-clonic seizures and moderate cognitive impairment.

Table 1. Summary of SCN1A Variants and Clinical Phenotypes

Patient	Gene	Transcript	cDNA	Protein	Origin	Zygosity	Phenotypes
SMEI-1	SCN1A	NM_001165963.4	c.3733C>T	p.Arg1245Ter	de_novo	heterozygous	Febrile seizures, myoclonic seizures, ataxia, sev
SMEI-2	SCN1A	NM_001165963.4	c.2216G>A	p.Arg739Gln	de_novo	heterozygous	Febrile status epilepticus, GTCS, moderate cogn

Functional_Assay

Whole-cell patch-clamp recordings in HEK293 cells expressing the mutant SCN1A p.Arg739Gln channel demonstrated a complete loss of functional sodium current density compared to wild-type NaV1.1 channels, confirming a deleterious loss-of-function mechanism.